

BiGLSG: BiGRU-LSTM and GAN-based Intrusion Detection System for the IoT Network

Mid-term Project Report

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Under the Supervision of
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DECLARATION

I hereby certify that the work, which is presented here in the report, entitled **BiGLSG: BiGRU-LSTM and GAN-based intrusion detection system for the IoT network**, in fulfillment of the requirement for the award of the Bachelor of Technology degree and submitted to the institution, is an authentic record of my work carried out during the period January 2025 to March 2025.

I further declare that I have been mentored by Dr. Saumya Bhadauria. I affirm that the information presented in this report is accurate and true to the best of my knowledge.

Signature of Mentor

Signature of candidate

ACKNOWLEDGEMENT

I want to express my sincere gratitude to my mentor, Dr. Saumya Bhadauria, for giving me the opportunity to work on this project under his guidance. She has taught me to conduct research with rigor, explain findings clearly, and approach complex tasks methodically. Working under her direction has been a privilege and a valuable learning experience. I am deeply thankful to everyone who has helped me understand these concepts and encouraged my exploration.

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Sakshi

ABSTRACT

Intrusion detection systems (IDS) face significant challenges in handling high-dimensional, large-scale, and imbalanced network traffic data. The widespread integration of IoT devices in smart agriculture (SA) further exacerbates security vulnerabilities as attackers exploit resource-constrained edge nodes and network gateways to launch sophisticated cyberattacks. To address these challenges, we propose a deep learning-based IDS that combines a denoising autoencoder (AE) with a Wasserstein Generative Adversarial Network (WGAN) to improve feature extraction and mitigate class imbalance. The AE captures compact, high-representative features, while the WGAN, enhanced with gradient penalty for stable training, generates realistic attack samples to enhance model robustness. For real-time threat detection, we employ a hybrid classifier integrating bidirectional gated recurrent units (BiGRU) and long short-term memory (LSTM) networks. The proposed IDS is deployed at the fog layer, enabling low-latency detection while reducing the computational burden on centralized cloud systems. The effectiveness of BiGLSG is validated using three benchmark datasets, BoT-IoT, Edge-IIoTset, and IoT-23, demonstrating superior performance in intrusion detection compared to existing methods.

Key Words

IoT: The Internet of Things

IDS: Intrusion detection System

SA: Smart agriculture

AE: Autoencoder

GAN: Generative Adversarial Network

WGAN: Wasserstein Generative Adversarial Network

GP: Gradient Penalty

BiGRU: Bidirectional gated recurrent units

LSTM: Long Short-Term Memory

DL: Deep learning

AIDS: Anomaly-based Intrusion Detection Systems

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CHAPTER 1

1. INTRODUCTION

1.1. Background

The Internet of Things (IoT) has transformed traditional network communication by enabling seamless connectivity among heterogeneous devices, facilitating significant advancements in smart agriculture (SA). By integrating information technologies, intelligent control, and data-driven decision-making, IoT enhances productivity and efficiency in various SA applications [1]. However, IoT deployment in SA faces environmental challenges such as wind, snowfall, flooding, and complex topography. Additionally, resource constraints, including low power consumption, limited computational capacity, and a lack of robust security infrastructure, make IoT devices susceptible to cyber intrusions (Sarjan, Ameli, & Ghafouri, 2022 [2]). These limitations hinder the implementation of strong security mechanisms, such as encryption algorithms, which require high computational power (Vishwakarma & Jain, 2020 [3]).

Intrusion detection systems (IDSs) play a crucial role in securing IoT networks by identifying cyber threats targeting IoT devices. Traditional IDSs rely on manually crafted rules, which are labor-intensive to develop and often ineffective against evolving threats. Deep learning (DL) has emerged as a promising alternative, leveraging advanced algorithms such as deep belief networks (DBNs) [4], convolutional neural networks (CNNs) [5], and recurrent neural networks (RNNs) [6] to detect complex attack patterns from raw network traffic. By utilizing large-scale data and continuously adapting to new threats, DL-based IDSs offer significantly improved detection accuracy over conventional methods [7].

1.2. Motivation

The performance of DL-based IDSs heavily relies on the availability of high-quality training data. While benign network traffic is abundant, many attack categories are significantly underrepresented, leading to severe class imbalance issues. Common approaches to address this problem include undersampling and oversampling techniques; however, undersampling

discards valuable data, and oversampling risks overfitting by duplicating existing samples. A more effective approach is synthetic data generation, which enhances model generalization and mitigates high false-negative rates.

The effectiveness of DL-based IDS models also depends on their ability to learn from diverse attack scenarios, which requires a balanced dataset that adequately represents all possible cyber threats. Given that new attack patterns continue to emerge, IDS models trained on imbalanced datasets may struggle to recognize novel threats. The use of generative adversarial networks (GANs) in IDS solutions allows for creating high-quality synthetic attack data, thus improving the model's ability to detect rare attacks while maintaining high detection accuracy for common threats.

Furthermore, IoT-based SA systems often operate in remote and resource-constrained environments where high latency, limited bandwidth, and unreliable network connectivity present significant challenges for cloud-based security solutions. In such settings, real-time threat detection and mitigation are critical, as delayed responses can result in severe consequences, including compromised agricultural infrastructure and disrupted operations. Traditional cloud-based IDS models introduce significant communication delays due to the time required for data transmission, processing, and response. These delays can make cloud-based IDS solutions unsuitable for mission-critical IoT applications in SA.

Fog computing emerges as a compelling solution by shifting computational processes closer to the data source, reducing the dependency on centralized cloud infrastructure. By deploying IDS models at the fog layer, security monitoring can be performed with lower latency, improving the responsiveness of threat detection mechanisms [8]. This approach also minimizes the bandwidth consumption of transmitting large volumes of raw network traffic to the cloud for analysis, making it a more efficient and scalable solution for IoT-based SA environments.

1.3. Project objectives

BiGLSG aims to design a robust IDS for IoT-based SA systems by integrating deep learning techniques and fog computing. The key objectives of this study are as follows:

1. Develop an IoT-Focused IDS::

- Propose an IDS that combines bidirectional gated recurrent unit (BiGRU) and long short-term memory (LSTM) networks with a softmax classifier to effectively detect cyber threats at the network edge.

2. Enhance Data Generation with AE-WGAN-GP::

- Utilize Autoencoder Wasserstein generative adversarial networks (AE-WGAN) with a gradient penalty to generate synthetic minority attack samples, addressing dataset imbalance and improving detection performance.

3. Implement a Fog-Based Deployment Architecture: :

- Propose a fog computing-based framework tailored to extreme agricultural environments, reducing reliance on cloud-based solutions and ensuring real-time threat detection.

4. Evaluate the Proposed Model::

- Assess the performance of the IDS using publicly available datasets, including BoT-IoT, Edge-IIoTSet, and IoT-23, which encompass a wide range of IoT-based cyber threats, ensuring a comprehensive evaluation.

This paper is organized as follows: Section II discusses the existing literature. The proposed methodology is described in Section III. In Section IV, we evaluate the performance of the proposed IDS and perform a comparison with state-of-the-art techniques. Finally, Section V concludes this work with future directions.

Chapter 2

2. LITERATURE REVIEW

This section reviews recent advancements in anomaly-based intrusion detection systems (AIDS), focusing on machine learning, deep learning, and generative models. It explores key developments in these approaches and their applications in addressing critical challenges in modern network security, including high-dimensional data processing, imbalanced dataset management, and zero-day attack detection.

2.1. Anomaly-based IDS

Various deep learning (DL) techniques have been explored to enhance intrusion detection in IoT networks. A DL-based anomaly detection mechanism proposed in [9] utilizes BiGRU, LSTM, and Truncated Backpropagation Through Time (TBPTT). Trained on CIC-IDS2018, ToN-IoT, and Edge-IIoT datasets, this model effectively detects multiple attack types, including DDoS, XSS, FTP-Patator, SSH-Patator, Botnet, MITM, Injection, Ransomware, and various subclasses of DoS attacks such as Goldeneye, Hulk, Slowloris, Portscan, DDoS_ICMP, DDoS_TCP, and Vulnerability Scanner.

Kandhro et al. [10] introduced a DL-driven vulnerability assessment framework incorporating DNN, CNN, and RNN classifiers. This model, trained on NSL-KDD, KDDCup99, and UNSW-NB15, demonstrates high performance in detecting Brute Force, DoS, Infiltration, and Botnet attacks. Similarly, a game-theory-based cloud security model (GT-CSDNN) in [11] was trained on CIC-IDS2018 and achieved significant accuracy in detecting cloud-based security threats.

A transfer learning approach in [12] utilized CNN-based models such as VGG16, VGG19, EfficientNets, Inception, and MobileNet to detect threats in cloud IoT environments. This model was trained on CIC-IDS2017 and CIC-IDS2018, proving effective for various types of cyber threats. Another study [13] applied federated learning concepts using CNN and MLP classifiers for intrusion detection in maritime transportation systems, demonstrating high efficacy on the NSL-KDD dataset.

Other approaches have focused on improving detection efficiency through different method-

ologies. A model integrating LightGBM and histogram-based gradient boosting (HBGB) [14] was trained on CIC-IDS2018, showing high efficiency in IoT attack detection. A deep autoencoder (DAE)-based model [15] was specifically designed for wireless spoofing and botnet attack detection, achieving a 98.6% detection accuracy.

A blockchain-assisted DL framework [16] incorporated deep variational autoencoders (DVAE) to enhance privacy preservation in Industrial IoT (IIoT) networks. This model was trained on ToN-IoT and IoT-Botnet datasets. In another study [17], DNN-based intrusion detection systems (IDS) demonstrated strong resilience against DoS and DDoS attacks in large-scale IoT communications.

Table 2.1 provides a comprehensive summary of these approaches.

Table 2.1: Comparison of Various IDS Models for IoT Security

Ref & Year	Approach	Targeted Attacks	Application Area	Classifier	Dataset
[9], 2024	A DL based Anomaly detection mechanism is designed.	DDoS, XSS, FTP-Patator, SSH-Patator, Botnet, MITM, Injection, Ransomware, Password, Injection, Backdoor, and DoS and its subclasses	Fog layer	BiGRU, LSTM, TBPTT	CIC-IDS2018, ToN-IoT, and Edge-IIoTset
[10], 2023	A vulnerability assessment framework is proposed.	Brute Force XSS, Brute Force WEB, DoS Hulk, DoS LOIC HTTP, Botnet attacks	IoT-enabled cyber-security systems	DNN, RNN, CNN	NSL-KDD, KDDCup99, UNSW-NB15
[11], 2022	A graph theory-based threat detection model is presented.	Cloud-based attacks	Cloud	DNN	CIC-IDS2018
[12], 2023	A transfer learning (CNN) approach is presented to detect intrusions.	DDoS, Web attacks, Infiltration attacks, Botnet attacks, Port scan attacks	Cloud IoT devices	CNN	CIC-IDS2017, CIC-IDS2018
[13], 2022	An intelligent adversaries investigation system is proposed.	Multifarious cyber attacks	IoT-based Maritime Transportation	CNN, MLP	NSL-KDD
[14], 2021	A novel ensemble attack monitoring framework is devised.	Bot, Brute Force attacks, Infiltration attacks, WEB attacks	IoT-empowered multi-attack classification environment	Light GBM, HBGB	CIC-IDS2018
[15], 2022	An anomaly detection mechanism is designed.	Wireless spoofing attacks, Botnet attacks	NGIoT	DAE	NA
[16], 2022	A blockchain-assisted threat detection framework is developed.	Data poisoning attacks, inference attacks	IIoT	DAVE, DGRNN	ToN-IoT, IoT-Botnet
[17], 2022	A novel threat detection model is presented.	Botnet attacks, DoS attacks	IoT-based smart communications	DNNGRU	CIC-IDS2018

2.2. GAN

With advancements in deep learning (DL), various models have been proposed for anomaly-based intrusion detection systems (AIDS) to process large-scale, high-dimensional, and imbalanced network traffic data. Generative adversarial networks (GANs) have been widely

employed to address data imbalance issues in AIDS.

Ieracitano et al. [18] combined autoencoders (AE) with statistical techniques, achieving F1 scores of 81.98% and 82.04% for binary and multi-class classification on the NSL-KDD dataset, outperforming traditional ML and DL models. Similarly, Lee and Park [19] enhanced AIDS performance by integrating AE with conditional GANs on CICIDS-2017, significantly improving rare attack detection.

Oguz Kaplan and Emre Alptekin [20] proposed a BiGAN-based one-class anomaly detection model for KDDCUP99; however, it lacked multi-class generalization. Furthermore, Zhang et al. [21] introduced CWGAN-CSSAE, which enhanced AIDS for detecting unseen and minority attacks on KDDTest-21 and UNSW-NB15.

Cui et al. [22] developed GMM-WGAN-IDS, integrating feature extraction, imbalance processing, and classification, achieving superior performance on NSL-KDD and UNSW-NB15 datasets.

Despite these advancements, limited work has explored AE with WGAN using gradient penalty, which provides more stable training than traditional GANs. This research gap motivates our AE-WGAN-GP approach, aiming to enhance synthetic attack generation and overall detection performance. Recent work by Gide and Mu'azu [23] demonstrated the effectiveness of hybrid models, achieving a 95.7% AUC in real-time DoS/DDoS detection on the MQTT-IoT-IDS2020 dataset. Their study aligns with our proposed AE-WGAN-GP-based IDS, emphasizing the potential of hybrid models in improving IoT security and real-time intrusion detection.

Table 2.2: Comparison of related work in GAN

Study	GAN Type	Dataset	Strength
[21]	CWGAN-CS	NSL-KDD	Improved minority attack detection
[19]	AE-CGAN	CICIDS-2017	Enhanced performance for rare attacks
[22]	GMM-WGAN-IDS	NSL-KDD, UNSW-NB15	Better handling of imbalanced data

Chapter 3

3. METHODOLOGY

This section outlines the proposed approach for applying the BiGLSG to intrusion detection tasks. The methodology is currently being explored and refined as the project progresses, with a focus on practical implementation within computational constraints.

3.1. Application Scenario

In a smart agriculture (SA) environment, IoT devices such as sensors, automated irrigation systems, and weather monitoring units often operate with default credentials or weak authentication mechanisms, making them highly susceptible to cyber threats. Additionally, these devices are frequently deployed in harsh and unattended environments, such as fields, greenhouses, or remote agricultural sites, where they are exposed to extreme weather conditions like heavy rainfall, storms, and extreme temperatures. This lack of physical security makes them vulnerable to node-capturing attacks, where an attacker gains control over an IoT device and injects malicious firmware or malware.

Once compromised, the infected IoT device becomes a part of a botnet network, operating under the control of a botmaster. The botmaster exploits system vulnerabilities to remotely manipulate compromised devices, coordinating them to execute cyberattacks. The compromised IoT devices communicate with the attacker's command-and-control (C2) server, receiving further instructions to launch large-scale attacks. Over time, this botnet expands, enhancing its capability to distribute malware across interconnected IoT nodes.

As shown in Figure 3.1, the intrusion detection framework processes real-time network traffic from IoT devices in the SA environment. It leverages data preprocessing techniques such as data cleaning, Z-score scaling, and one-hot encoding to structure network traffic data before feeding it into the detection model. The Bi-GRU and LSTM-based deep learning module analyzes sequential traffic patterns to detect malicious activity. Additionally, a GAN-based model enhances detection performance by generating synthetic attack traffic, mitigating class imbalance issues and improving anomaly detection.

Upon detecting an intrusion, the system classifies it as a DDoS, botnet, or other attack type

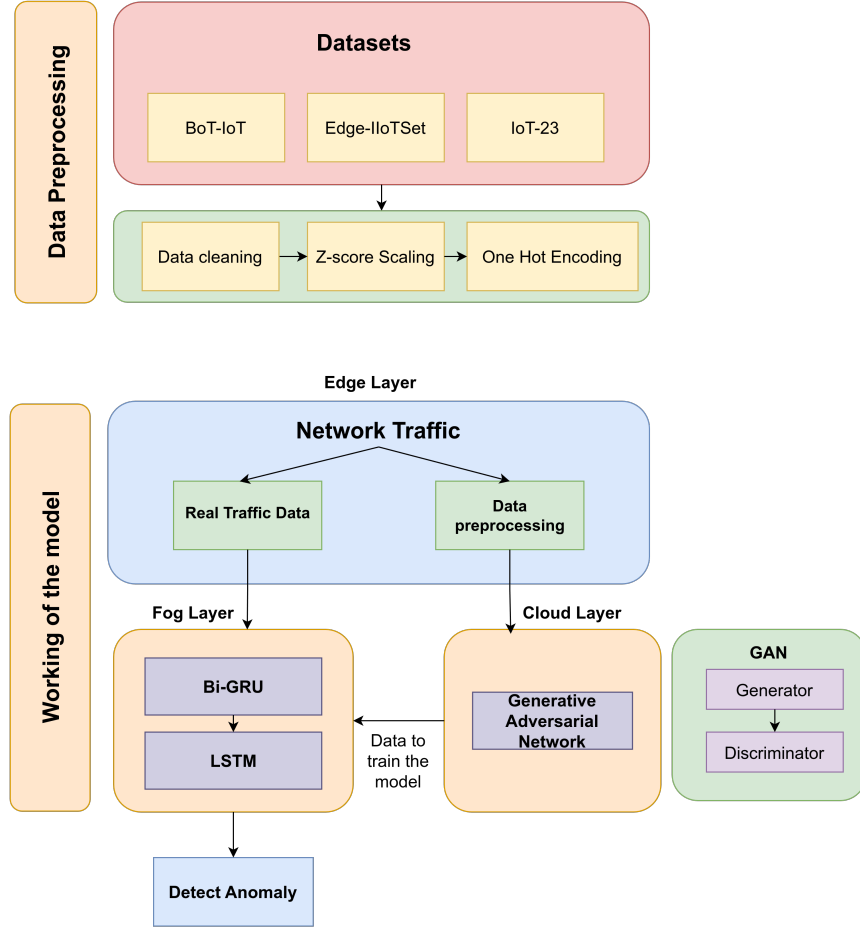


Figure 3.1: Deployment architecture for proposed IDS in the extreme environment of SA.

and issues an alert, enabling proactive security measures. Attackers frequently leverage bot-nets to launch DDoS attacks against edge servers, disrupting data aggregation, decision-making, and automated agricultural operations. The proposed IDS framework effectively mitigates such threats by providing real-time anomaly detection and enhanced network security in SA environments, safeguarding agricultural IoT infrastructure from malicious exploitation.

3.2. Data Preprocessing

Preprocessing data were conducted on each dataset where the size of the training set, validation set, and testing set were (70%, 20%, 10%) respectively in our experiments.

- **Data cleaning**

It is a crucial step in preparing raw datasets for training deep learning models. We have handled missing values by imputing them using the mean. Additionally, duplicate

entries are identified and removed to prevent overfitting, and inconsistent or erroneous data points are corrected to ensure data integrity.

- **Encoding**

It is essential when working with categorical features, as deep learning models require numerical inputs. One-hot encoding is used for nominal variables, while ordinal encoding is applied to features with a clear hierarchical relationship. In network intrusion detection, categorical attributes like protocol types, attack categories, and service types are transformed into numerical representations.

- **Scaling**

It ensures that numerical features are standardized or normalized to prevent models from being biased toward features with larger numerical ranges. Z-score normalization is used to transform features to have zero mean and unit variance. Scaling is particularly important for distance-based models and deep learning architectures that rely on gradient-based optimization.

- **Feature selection** It improves model efficiency and performance by eliminating irrelevant or redundant features. The Analysis of Variance method (ANOVA) was applied as a feature selection to select the most relevant features, which outperformed compared to Spearman's rank, Kendall's rank, and Pearson correlation techniques.

3.3. Background of GAN

- **3.3.1. Overview of basic GAN architecture**

Generative Adversarial Networks (GANs), introduced by Goodfellow et al. [24], generate high-quality synthetic samples by learning the data distribution. A GAN consists of two neural networks, as shown in Figure 3.2, the Generator (G) and the Discriminator (D). The Generator learns to create realistic data samples from random noise (Z), forming a generated distribution, $P(G)$, while the Discriminator distinguishes between real and generated samples. It assigns a probability ($P = 1$ for real, $P = 0$ for fake). The training process iteratively improves both networks, where the Generator loss (G-loss) measures the discrepancy between generated and real data, and the Discriminator loss (D-loss) quantifies misclassification errors.

Overall, the GAN framework involves a min-max game, as defined by Eq. (1), where P_g represents a distribution of data produced by the generator, and P_r denotes the real

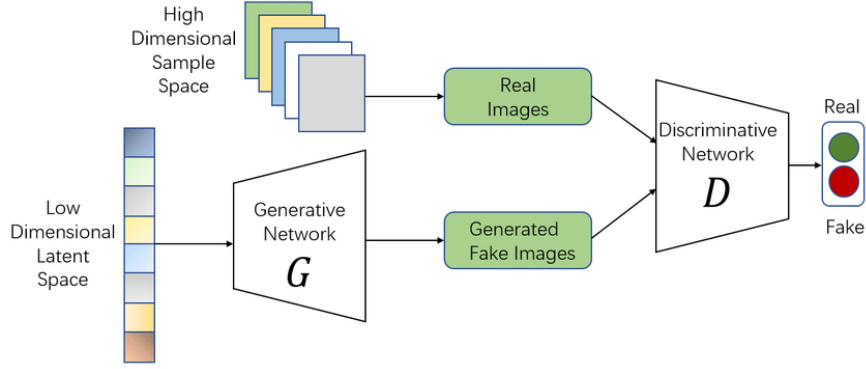


Figure 3.2: Standard GAN

data distribution.

$$\min_G \max_D V(D, G) = E_{x \sim P_r} [\log D(x)] + E_{z \sim P_g} [\log(1 - D(G(z)))] \quad (1)$$

• 3.3.2. BiGAN

A Bidirectional Generative Adversarial Network (BiGAN) [25] extends the traditional GAN by incorporating an Encoder (E) alongside the Generator (G) and Discriminator (D) as shown in Figure 3.3. The Encoder maps input features from high-dimensional space into a latent representation, enabling inverse feature learning. The Discriminator then classifies both real attack samples encoded by the Encoder ($E(x)$) and synthetic attacks generated by the Generator ($G(x)$). This joint distribution learning enhances feature extraction, making BiGAN effective for anomaly detection. The model ensures a bidirectional mapping between input and latent spaces, improving the quality of adversarial samples and aiding in robust attack detection.

The first objective in BiGAN architecture is to minimize the Generator (Fake Image (F_Image1)) and encoder (Real Image (R_Image1)) loss. While the second objective is to maximize the Discriminator loss. The BiGAN objectives can be represented in Eqs. (1) and (2), respectively.

$$\min G_1(z, \theta_{G1}), \quad E_1(x, \theta_{E1}) \quad (1)$$

$$\max D_1(\{x, z\}; \theta_{D1}) \quad (2)$$

The minimax equation in BiGAN contains three components: one discriminator (D),

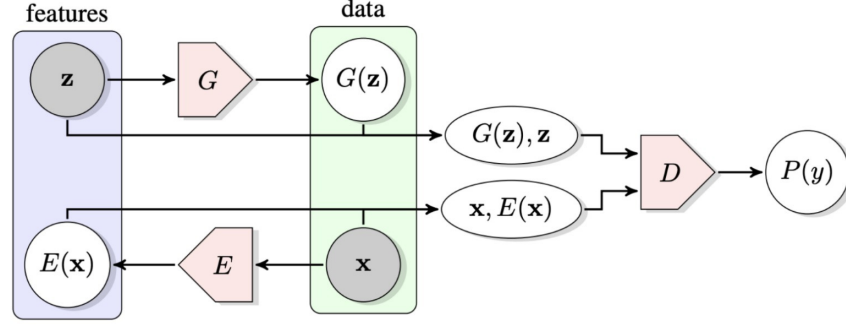


Figure 3.3: Bidirectional GAN

one generator (G), and one encoder (E), which are different from the minimax game of GANs, as in vanilla GANs, the Discriminator tries to discriminate between the real and adversarial samples, as in Eqs. (3) and (4), respectively.

$$D_{\text{real}} = D(\{x, E(x)\}; \theta_D) \quad (3)$$

$$D_{\text{fake}} = D(\{G(z; \theta_G), z\}; \theta_D) \quad (4)$$

According to the minimax game equation, the discriminator between real and fake samples (D) can be substituted in Eq. (5), as indicated in Eq. (6).

$$\min_{\theta_G} \max_{\theta_D} V(G, D) = \min_G \max_D E_{x \sim P_{\text{data}}} [\log D(x)] + E_{z \sim P_z} [\log(1 - D(G(z)))] \quad (5)$$

$$V(D, G, E) = \log D(x, E(x, z); \theta_D) + \log(1 - D(\{G(z; \theta_G), z\}; \theta_D)) \quad (6)$$

Eq. (6) is defined to work on a single sample input, but for multiple samples of traffic, the expectation (mean) is included, as follows in Eq. (7).

$$\begin{aligned} V(D, G, E) = & E_{x \sim p_x(x)} E_{z \sim p(z|x)} [\log(D(\{x, E(x, z)\}; \theta_D))] \\ & + E_{z \sim p(z)} E_{x \sim p(x|z)} [\log(1 - D(\{G(z; \theta_G), z\}; \theta_D))] \end{aligned} \quad (7)$$

• 3.3.3. AE-WGAN-GP

The bidirectional structure of BiGAN offers key advantages for anomaly detection in network security. First, it enables simultaneous learning of data generation and inverse

mapping, facilitating comprehensive feature representation. Second, the encoder effectively captures normal network traffic distributions. Third, the learned latent space provides an efficient mechanism for detecting behavioral deviations.

However, traditional BiGANs face limitations in generating diverse attack samples due to mode collapse, where the generator produces only a restricted subset of patterns. This is critical for IDS, which requires diverse attack representations. To address this, Arjovsky et al. [26] introduced WGAN, which optimizes training stability by using Wasserstein distance to measure distribution differences (P_r, P_g), reducing mode collapse and improving computational efficiency.

Wasserstein GAN with Gradient Penalty (WGAN-GP) refines the original WGAN by replacing weight clipping with a gradient penalty (GP) technique as shown in Figure 3.4. Weight clipping can cause capacity issues and requires additional parameters to define permissible weight spaces, whereas GP mitigates these challenges by enforcing the 1-Lipschitz constraint more effectively.

Regularization techniques in WGAN-GP

1. The discriminator replaces batch normalization with layer normalization to improve stability.
2. Spectral Normalization (SN) (Miyato et al., 2018) ensures that the discriminator (D) maintains K-Lipschitz continuity by progressively constraining the Lipschitz constant of each layer.
3. Gradient Penalty (GP) allows for more frequent training with minimal hyperparameter tuning. Instead of clipping, GP penalizes deviations from the 1-Lipschitz constraint, adjusting the penalty based on how far the gradient norm deviates from 1.

The WGAN-GP loss combines the original critic loss with the gradient penalty loss, defined as follows:

$$L_{\text{Critic}} = E_{x \sim P_r}[D(x)] - E_{x \sim P_g}[D(G(x))]$$

$$L_{\text{GP}} = \lambda \cdot E_{\hat{x} \sim P_{\hat{x}}} [(\|\nabla_{\hat{x}} D(\hat{x})\| - 1)^2]$$

The gradient norm for random samples $\hat{x} \sim P_{\hat{x}}$, between pairs of points sampled from the data distribution P_r and the generator distribution P_g , $P_{\hat{x}}$ sampling is conducted uniformly along straight lines. The regularization coefficient λ has been set to 10. This has been empirically proven to work well with a wide range of architectures and datasets.

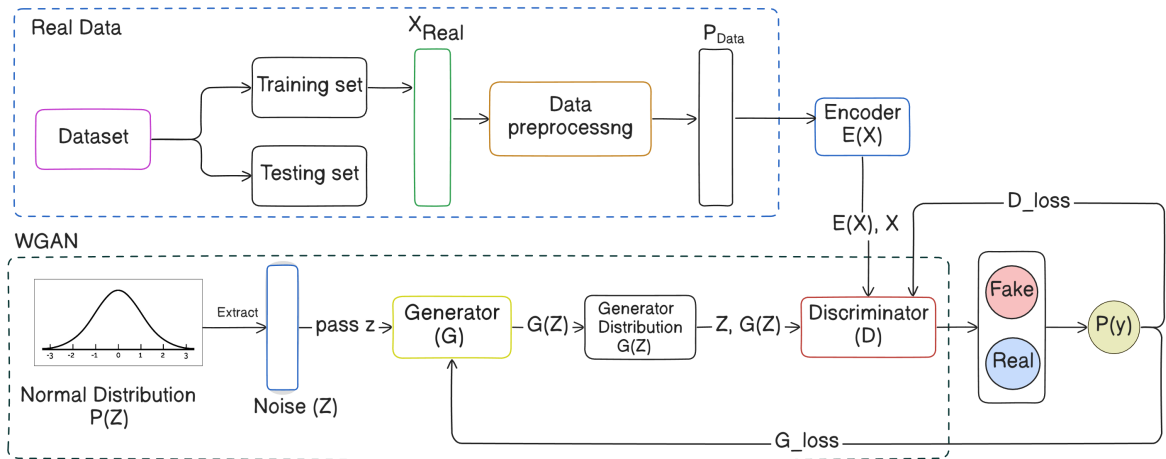


Figure 3.4: AE-WGAN-GP

Chapter 4

4. PERFORMANCE EVALUATION

1. 4.1. Experimental Setup

The proposed IDS is implemented on a high-performance computing setup featuring a Lenovo Legion PC with 64 GB RAM and an Intel Core i9-13900K processor operating at 3.50 GHz. The system is equipped with an NVIDIA GeForce RTX 4090 GPU with 24 GB VRAM to accelerate deep learning model training and inference. Additionally, the implementation leverages Python-based deep learning libraries, including TensorFlow, Keras, NumPy, Pandas, and Scikit-learn, to facilitate efficient data pre-processing, model training, and evaluation.

2. 4.2. Datasets

- **BoT-IoT**

The Bot-IoT dataset (Koroniotis et al., 2019) is chosen because it represents the environment where the IoT is attacked by DoS and botnet. It includes normal IoT network traffic and many kinds of attacks, which can be applied to evaluate the system proposed in this paper. There are 5 types of attacks, namely DDoS, DoS, OS/Service Scan, Keylogging, and data exfiltration attacks. Therefore, this experiment selects all the data of Bot-IoT to evaluate the HDA-IDS. The captured pcap files are 69.3 GB in size, with more than 72.000.000 records. To ease the handling of the dataset, we have used 5% of the original dataset, which comprises 4 files of approximately 1.07 GB total size and about 3 million records.

- **Edge-IIoT**

The Edge-IIoTset dataset is a benchmark dataset for intrusion detection in industrial IoT (IIoT) and edge computing environments. Developed using realistic IIoT testbed, it simulates industrial applications like smart grids and intelligent transportation. The dataset includes benign and malicious traffic, covering attack types such as Denial of Service (DoS), Man-in-the-Middle (MitM), botnet, ransomware, and malware-based attacks. It features high-dimensional network traffic data, enabling the evaluation of AI-driven IDS models. Edge-IIoT is widely used for developing and benchmarking intrusion detection systems, addressing

real-world IIoT security challenges in resource-constrained, latency-sensitive industrial networks.

- **IoT-23**

The IoT-23 dataset is a comprehensive benchmark dataset for malware and intrusion detection in IoT environments. Created by Stratosphere Laboratory, it contains real-world network traffic captured from both benign IoT devices and malware-infected devices. The dataset includes 23 scenarios, covering various attack types such as DDoS, data exfiltration, scanning, and command-and-control (C2) communications. It provides labeled traffic data, allowing researchers to develop and evaluate intrusion detection models. IoT-23 is widely used for studying IoT security threats, improving anomaly detection techniques, and enhancing cybersecurity solutions for resource-constrained IoT networks.

4.3. Evaluation Metrics

All the standard and extended metrics of evaluation, i.e., accuracy (ACC), recall (REC), F1-score (F1), precision (PRE), true positive rate (TPR), confusion matrix (CNM), false positive rate (FPR), ROC curve, true negative rate (TNR), false discovery rate (FDR), Mathhews correlation coefficient (MCC), false omission rate (FOR), and false negative rate (FNR), are used to thoroughly evaluate the performance of the proposed IDS. The following equations compute these metrics:

$$ACC = \frac{TPR + TNR}{TPR + TNR + FPR + FNR} \quad (4.1)$$

$$REC = \frac{TP}{TPR + FNR} \quad (4.2)$$

$$F1 = 2 \times \frac{PRE \times REC}{PRE + REC} \quad (4.3)$$

$$PRE = \frac{TPR}{TPR + FPR}. \quad (4.4)$$

Chapter 5

5. CONCLUSION

This study presents a deep learning-based intrusion detection system (IDS) specifically designed to enhance security in IoT-driven smart agriculture environments. As IoT devices become increasingly integrated into smart farming systems, they face growing risks from sophisticated cyber threats, necessitating advanced security mechanisms. The BiGLSG IDS leverages the power of generative adversarial networks (GANs) to address key challenges such as data imbalance and feature extraction inefficiencies in intrusion detection.

The model employs a hybrid approach, integrating a denoising autoencoder (AE) with a Wasserstein GAN (WGAN) to generate high-quality synthetic attack samples. This data augmentation technique mitigates class imbalance issues commonly found in cybersecurity datasets, ensuring the IDS learns from diverse attack patterns, including rare and emerging threats. Additionally, the feature extraction process is significantly improved, allowing for a more discriminative and representative dataset that enhances detection accuracy.

For real-time attack detection, a hybrid BiGRU-LSTM classifier is deployed at the fog computing layer, where security processing occurs closer to the data source. This reduces dependency on centralized cloud systems, leading to lower latency, faster response times, and decreased bandwidth consumption. By utilizing bidirectional gated recurrent units (BiGRU) alongside long short-term memory (LSTM) networks, the classifier efficiently captures both sequential dependencies and temporal correlations within network traffic data, significantly enhancing intrusion detection performance.

To validate the effectiveness of the proposed IDS, extensive experiments were conducted using three widely recognized cybersecurity datasets: BoT-IoT, Edge-IIoTset, and IoT-23. These datasets encompass a broad range of IoT-based cyber threats, including DDoS attacks, botnets, data poisoning, man-in-the-middle attacks, and various adversarial exploits. Comparative evaluations against state-of-the-art IDS approaches demonstrate that the proposed model achieves higher detection accuracy, lower false-positive rates, and greater robustness against evolving attack patterns. The results further underscore the impact of GAN-based synthetic data generation in improving IDS performance, particularly in scenarios where real-world attack data is sparse or imbalanced.

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